



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAS (24AAS3101)

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TITLE OF THE PROJECT		
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Abstract

AI-Based Driver Assistance System Using Computer Vision and Deep Learning for Real-Time Road Safety is designed to improve road safety by assisting drivers in identifying potential hazards and making safer driving decisions. The proposed system continuously analyses camera or video input using Computer Vision and Deep Learning techniques to detect important road elements such as vehicles, pedestrians, obstacles, traffic signs, and road lanes. It further analyses the detected objects, lane position, and approximate distance to identify potentially dangerous situations such as lane departure, unsafe proximity, and possible collision risks.

When a hazardous situation is detected, the system generates appropriate audio or visual alerts to warn the driver in real time. The system follows a processing pipeline of camera/video input, object and lane detection, risk analysis, danger identification, and driver alert generation.

The proposed system focuses on reducing accidents caused by human error and improving driver awareness through timely safety warnings. It can be implemented and evaluated using recorded road videos or a simulated environment, allowing its performance to be tested across different road and traffic conditions. The system will be evaluated based on factors such as detection accuracy and response time. Overall, the project aims to provide a practical AI-based driver assistance prototype that contributes to safer, more intelligent, and reliable driving.

Signature of the Guide

HOD